

Plan is managed by an independent crown corporation that can set its own salaries and hire whomever it likes. It is designed to be as far removed as possible from meddling politicians.⁴⁸ Norway's SWF is managed by a separate division of its central bank, a structure that allows it to operate somewhat independently of the rest of the institution.

Intergenerational Equity

National pension plans and defined benefit plans ensure retirement security by having different generations pay and receive different benefits. Social security pensions paid to one generation must be financed by generations following them, and private defined pension plans must be paid by a paternalistic company making contributions over time.

In a sustainable pension plan, the sum of all generational accounts must be zero: a benefit enjoyed by one generation must be paid for by another generation. Blake (2006) reports that there is large inequality across generations in the Social Security system. The generation born in the 1920s and 1930s is “the most favored generation in history according to the generational accounts” and the least favored generation is the cohort of the youngest workers today.

Some form of intergenerational inequality is actually optimal. *Intergenerational risk sharing* allows Social Security to absorb a shock that can be catastrophic for one generation—like those who were in the peak earnings years when the Great Depression struck—and diversifies it across a series of generations.⁴⁹ In defined benefit plans, employers help smooth risk across cohorts of workers. The amount of inequality in many pension systems today is very large and disadvantages younger workers; the old have stolen from the young.

The decline in the *dependency* (or *support*) *ratio*, which is the ratio of current workers to retirees, makes this problem worse. As people live longer, more retirees are being supported by fewer workers. The dependency ratio shrank from 5.3 in 1970 to 4.6 in 2010 and is projected to hit 2.6 in 2050.⁵⁰ The large retiree population in General Motors contributed to its bankruptcy: before bankruptcy it was supporting 2.5 retirees for every active worker, and retirees accounted for 70% of its health costs.⁵¹

Norway, like many developed countries, has an aging population and needs to save to meet increasing retiree costs. Between 2010 and 2060, the number of

⁴⁸ See “Factor Investing: The Reference Portfolio and Canada Pension Plan Investment Board,” Columbia CaseWorks ID#120302.

⁴⁹ Intergenerational accounting was introduced by Auerbach, Gokhale, and Kotlikoff (1991) and uses the concept of successive *overlapping generations*, with each generation following a life-cycle model (see chapter 5). Paul Samuelson (1958) and Peter Diamond (1965) developed overlapping generations models, which can be used to measure generational inequity.

⁵⁰ As reported in “Falling Short,” *Economist*, April 9, 2011.

⁵¹ Hakim, D., and J. W. Peters, “For GM Retirees, A Growing Sense of Unease,” *New York Times*, June 30, 2005.